

EMPLOYEE DATA ANALYSIS

Google Data Analytics Certificate

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PROJECT OBJECTIVE

Analyze employee data to understand the causes of employee turnover and identify factors influencing employee retention or departure, using the six-phase Google Data Analytics methodology

(Ask, Prepare, Process, Analyze, Share, Act).

DATA SOURCE :

Kaggle - Employee/HR Dataset

Link: [Employee/HR Dataset \(All in One\)](#)

Four main files were integrated:

- Employee Data : 3,000 employees, 27 columns
- Training Data : 3,000 records, 9 columns
- Employee Engagement Survey Data : 3,000 surveys, 5 columns
- Recruitment Data : 3,000 applicants, 20 columns



DATA QUALITY DISCOVERY

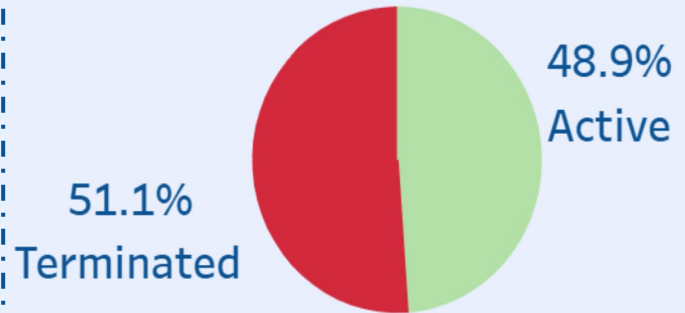
The original Employee Status column was found to be inconsistent:

- **38% of data** (1,146 employees) was **misclassified**.
- Active status incorrectly included employees who had already left.
- Future Start and Leave of Absence were completely wrong (100% error).

ACTION TAKEN:

- Created a new column "**Status_Corrected**" based solely on Start Date and Exit Date.
- **TRUE TURNOVER RATE** After correction:
51.1% of employees have left the company (1,533 Terminated out of 3,000 total employees).

Rotation rate



Overview

Total Employees
3,000

Active Employees
1,467

Turnover Rate
51.1%

Training Completion
50.30%

WHAT DID WE DISCOVER AFTER CLEANING THE DATA?

After fixing 38% misclassified employee records, the truth emerged:

- 📌 More than **HALF** of employees left the company (51.1%)
- 📌 Only 48.9% of employees remain active

These numbers tell a concerning story. But the real question is:

WHY are employees leaving?

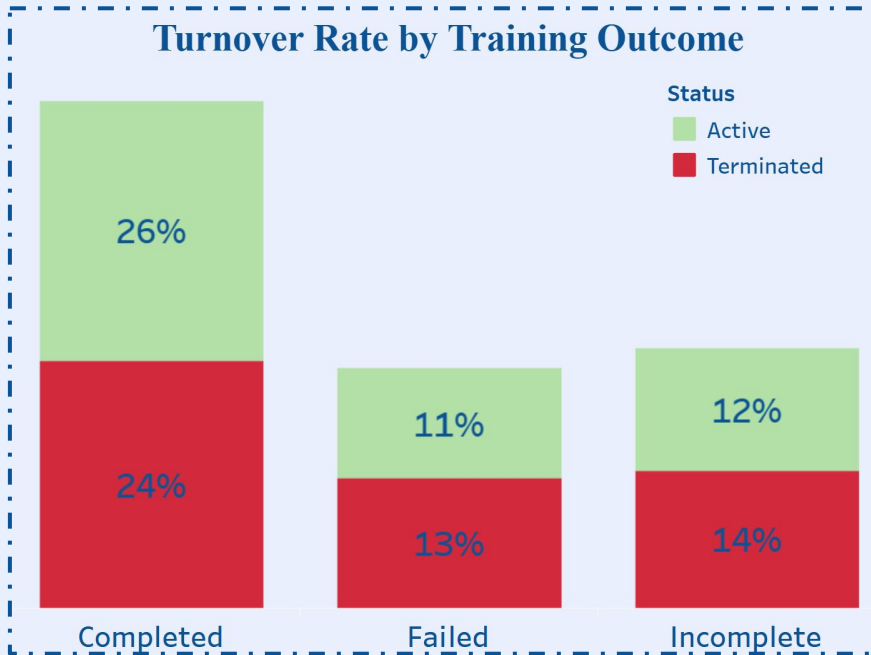
This analysis dives deep into three key areas:

- 1] Training Impact – Does training make employees stay?
- 2] Department Differences – Which teams lose the most people?
- 3] Termination Reasons – Is it voluntary, retirement, or involuntary?

Let's explore the answers in the following dashboards.



Training Impact on Turnover



THE SURPRISING TRUTH ABOUT TRAINING

Does training actually help keep employees?

The data gives us a clear answer:

 FAILED Training → **54.3%** left the company

 INCOMPLETE Training → **52.9%** left the company

 COMPLETED Training → **48.6%** left the company

THE GAP: Employees who FAIL training are 5.7% MORE likely to leave

than those who COMPLETE it.

WHAT THIS MEANS:

Training isn't just about learning new skills.

It's about engagement, confidence, and feeling valued.

When employees struggle and fail in training, they don't just fail a course.

They start questioning their place in the company.

BOTTOM LINE:

Fixing training programs isn't an HR expense.

It's a retention strategy.

Termination Reasons by Department

WHY DO EMPLOYEES LEAVE? IT DEPENDS ON THE DEPARTMENT.

Not all departures are the same. Each department tells a different story.

🔴 **SOFTWARE ENGINEERING: 60% Voluntary**

The highest voluntary turnover. Engineers are choosing to leave.

Question: Are we failing to engage our tech talent?

🟡 **SALES: 37% Retirement**

The highest retirement rate.

A wave of experience is walking out the door.

Question: Are we ready for the knowledge gap?

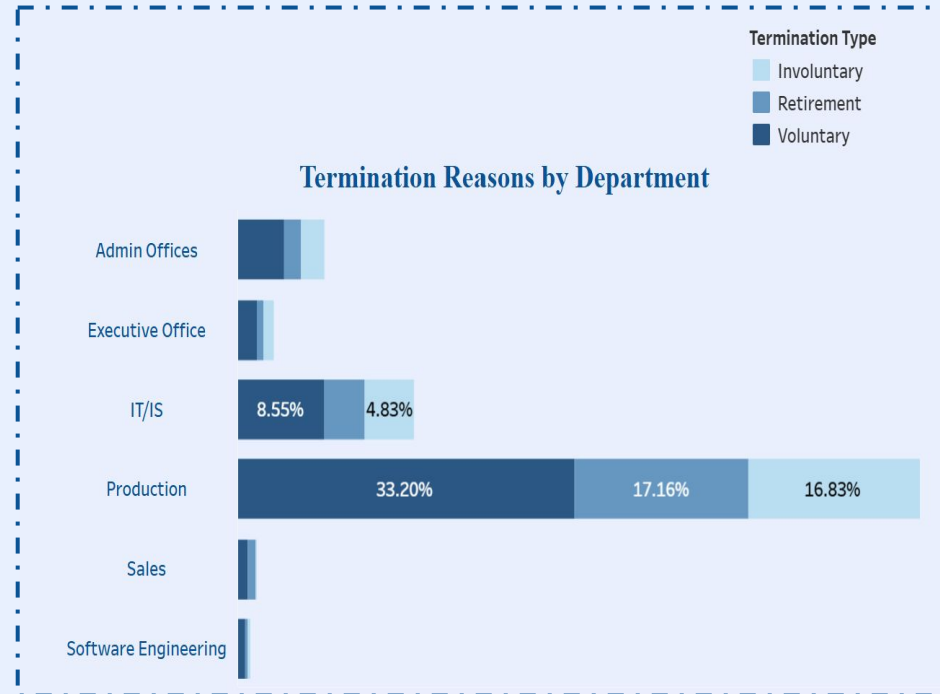
🟢 **PRODUCTION: Balanced (25% each)**

The most stable department. Voluntary, retirement, and involuntary reasons are equal.

WHAT THIS MEANS FOR YOU:

One retention strategy does **NOT** fit all departments.

- Software Engineering needs engagement and career growth.
- Sales needs succession planning and knowledge transfer.
- Production needs to maintain its balanced approach.



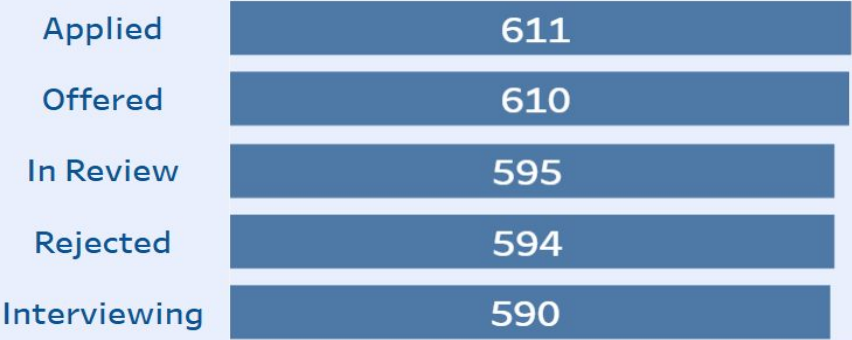
BOTTOM LINE:

Understand why each department loses people.

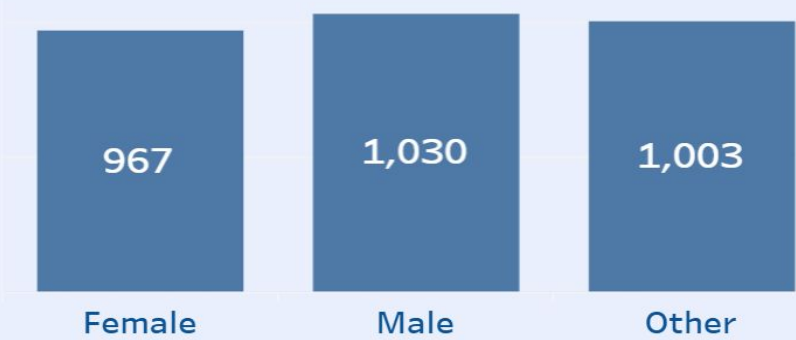
Then build department-specific retention strategies.

Recruitment Dashboard

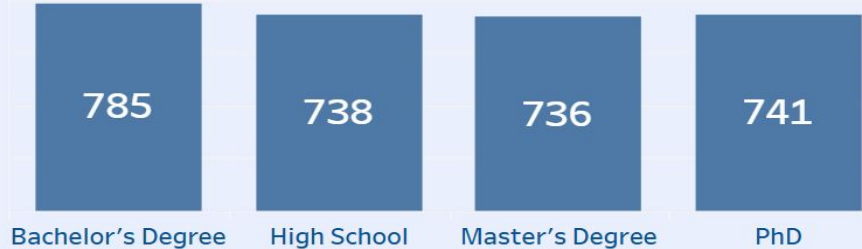
Recruitment Status



Recruitment Gender



Recruitment Education



Recruitment Monthly



Recommendations

Based on the analysis, here are 5 actionable recommendations:

1. IMPROVE TRAINING PROGRAMS

- Employees who fail training are **5.7%** more likely to leave
- Redesign training for Admin Offices and IT/IS
- Target: Reduce failure rate from **25%** to below **15%**

2. FOCUS ON SOFTWARE ENGINEERING RETENTION

- **60%** of departures are voluntary
- Conduct exit interviews, improve work environment
- Target: Reduce voluntary turnover to **40%**

3. SUCCESSION PLANNING FOR SALES

- **37%** of departures are due to retirement
- Implement knowledge transfer program
- Train new employees before retirement wave hits

4. CONDUCT SYSTEMATIC EXIT INTERVIEWS

- Current data doesn't explain **why** employees leave
- **Add** qualitative questions to understand real reasons

5. AUTOMATE DATA QUALITY CHECKS

- **38%** of employee status data was incorrect
- Implement periodic data validation



Conclusion

PROJECT SUMMARY:

- 4 files analyzed: Employee, Training, Survey, Recruitment
- **38% data error** discovered and corrected
- True turnover rate: 51.1% (not 12.9%)

KEY FINDINGS:

1. Training outcome is the strongest predictor of turnover
→ **Failed: 54.3% left** | **Completed: 48.6% left**
2. Departments need different retention strategies
→ **Software Engineering: 60% voluntary**
→ **Sales: 37% retirement**
3. Age, gender, and performance are **NOT** influencing factors

LIMITATIONS:

- This is synthetic data, not real-world organizational data

NEXT STEPS:

- ✓ Prioritize training improvement
- ✓ Implement retention strategies for Software Engineering
- ✓ Begin succession planning for Sales
- ✓ Automate data quality checks



REPORT Employee Analysis



Thank You

For reviewing this project.

 Interactive Tableau Story:
LINK